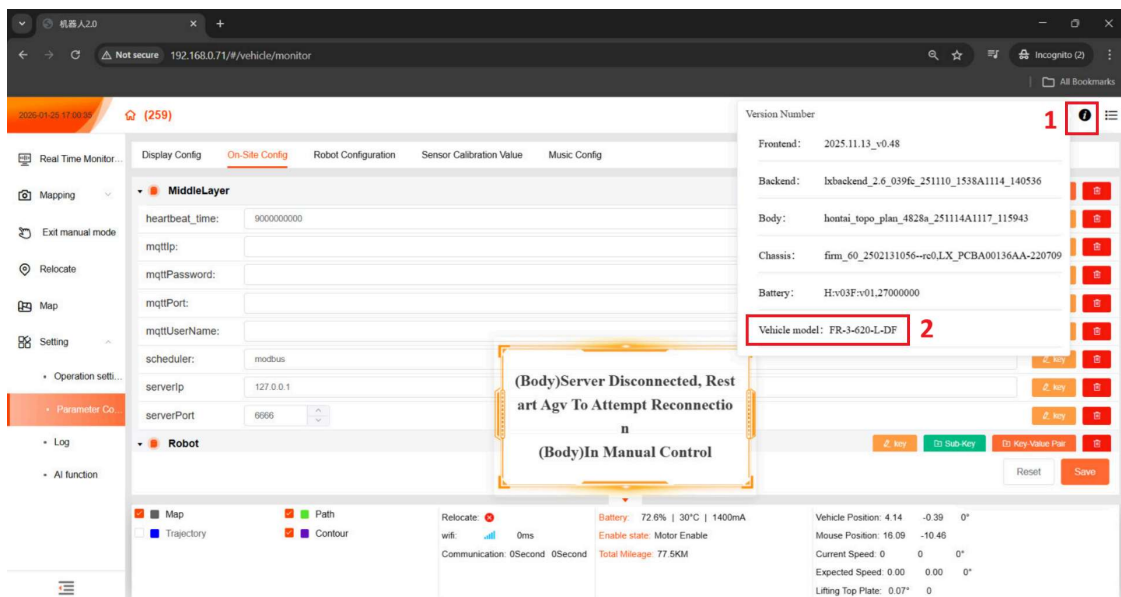


Lanxin Integration Handbook

Lanxin Controller

1. Open browser of Lanxin controller (<http://<NavWiz IP Address>:9998>). Screenshot the **Version Number** and upload it to google share drive (project or production) with folder name of AMR serial number. Get the **Vehicle model**
 - **FR-3-400-L-xx**
 - **FR-3-620-L-xx**
 - **FR-3-1010-L-xx**

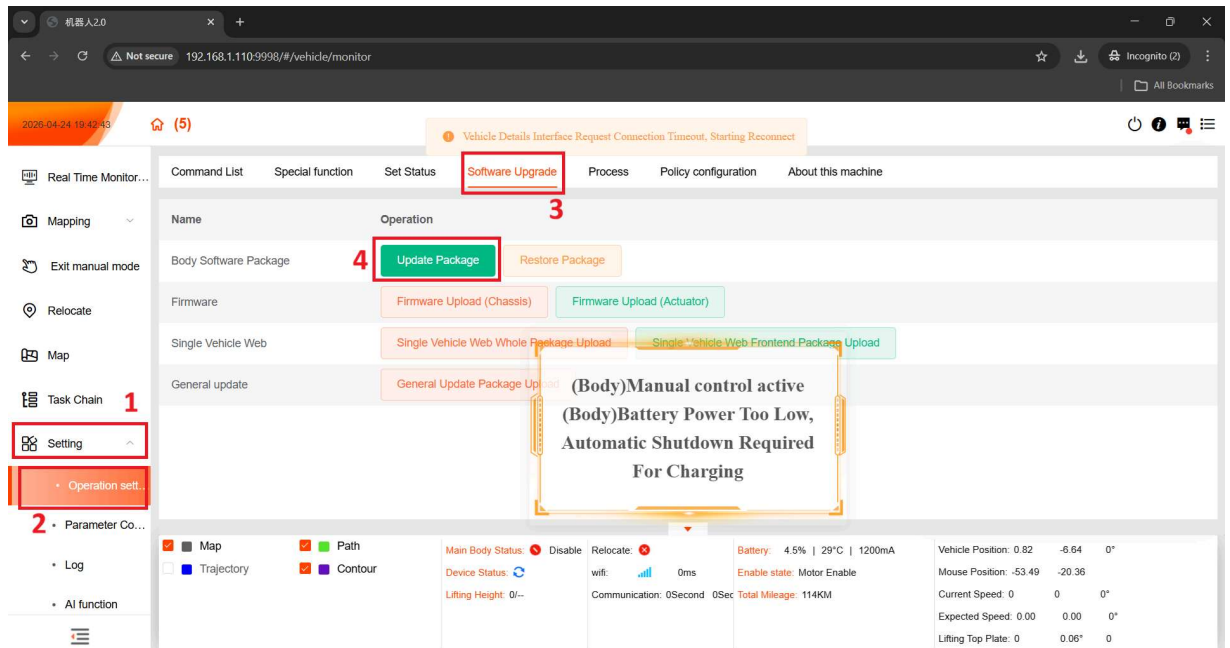


2. FileZilla **SFTP** login Lanxin controller.

```
Host: sftp://<NavWiz IP Address>
Username: fr1511b
Password: admin1234
Port: 20122
```

3. FileZilla download **~/hontai/params/config_*.ini** and upload to google share drive (project or production) with folder name of AMR serial number same as in step #1.

- Open browser of Lanxin controller (<http://<NavWiz IP Address>:9998>) and upload Body Software Package [hontai_topo_plan_4828a_251114A1117_115943_md5e6105.zip](#)



- Edit the downloaded **config_FR3xxxxLxx.ini** (matched with **Vehicle model**) in step #3.
 - Under **Ability** add/set **safe_manual_control=false**.
 - If the **port=9201** is missing under any **[Comm<#>]**, add **new Comm number <#>** (other than the existing number).

```
;AD不带套筒
[Ability]
open_slip_reset=true
have_manual_button=false
safe_manual_control=false
; manual_max_vx=0.5
; manual_max_vy=0.0
; #单位度/s
; manual_max_w=30

;modbus tcp节点
[Comm<#>]
host=any
mode=2
pattern=4
port=9201
protocol=18
type=7
```

6. FileZilla upload the edited file in step #5 to **Overwrite** `~/hontai/params/config_*.ini` (matched with **Vehicle model**). Make sure **file attributes** are **NOT changed**.
7. FileZilla download `~/hontai/start.sh` and upload it to google share drive (project or production) with folder name of AMR serial number same as in step #1.
8. Edit the downloaded **start.sh** in step #7.
 - Replace **"FR-3-620-L-DF"** (in line 199) with the current **Vehicle model**.

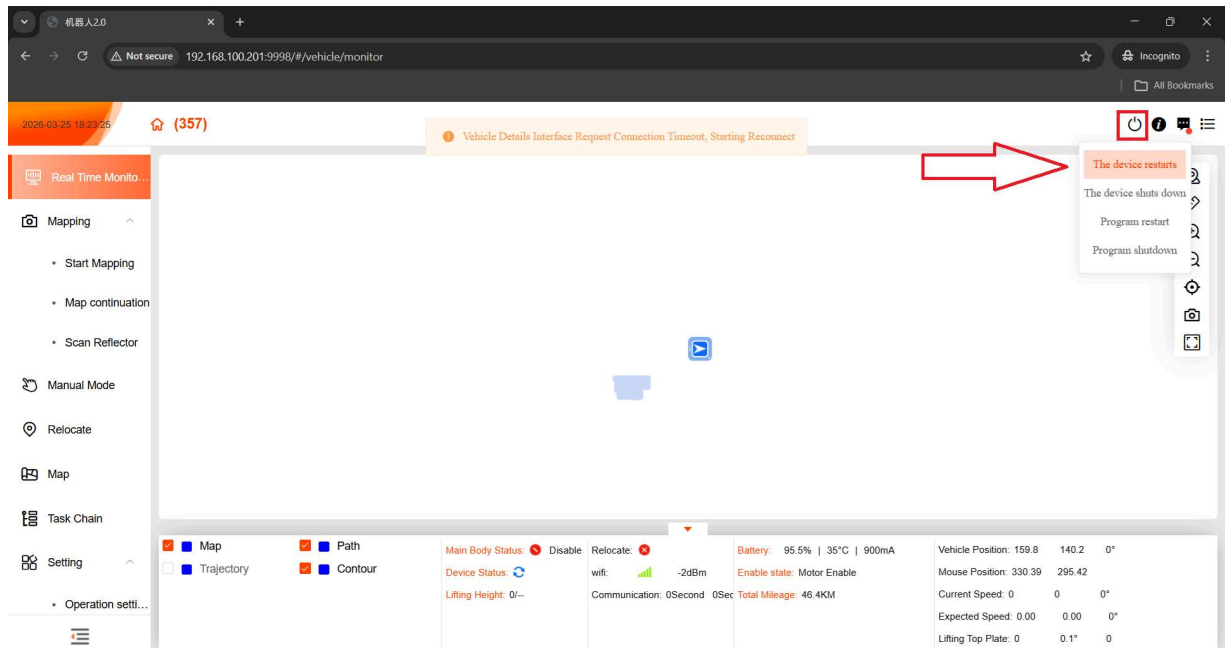
```

root@f151b@localhost: /hntal/ # ./f151b_enable_module.sh /hntal/perams/config_ate.ini | awk -F= '{print $2}'
if [ "$enable_module" != "" ]; then
    echo "配置指定模块名称 $enable_module"
    IFS=$'\n'
    read -ra modules << "$enable_module"
    ./ixRobot --with DataCenter Logger $${modules[@]} &
elif [ ${arr[@]:0:1} != "" ]; then
    echo "配置设备"
    ./ixRobot --with DataCenter Logger Driver Dsl Odom DeviceDriver Docking DockLocalization $${LocalizationModuleName} TrackNavi ObstacleDetectorV2 CloudProcessor Comm $${SlamModuleName}
    Laser Camera QRDetector DepthCamera $${DepthSemanticModuleName} MiddleController DefaultTask AudioController Calibration VisionBag MemCatch ImuFusion Imu &
elif [ ${arr[@]:0:1} != "SR" ] && [ ${arr[@]:0:1} != "L" ]; then
    echo "配置相机"
    ./ixRobot --with DataCenter Logger Driver Dsl Odom DeviceDriver Docking DockLocalization $${LocalizationModuleName} TrackNavi ObstacleDetectorV2 CloudProcessor Comm $${SlamModuleName}
    Laser Camera QRDetector DepthCamera $${DepthSemanticModuleName} MiddleController DefaultTask AudioController Calibration CoilAxisLocator MemCatch ImuFusion Imu &
elif [ "${robot_type}" = "SR-S-360-S-A2" ] ]; then
    echo "配置激光雷达"
    ./ixRobot --with DataCenter Logger Driver Dsl Odom DeviceDriver Docking DockLocalization $${LocalizationModuleName} TrackNavi ObstacleDetectorV2 CloudProcessor Comm $${SlamModuleName}
    Laser Camera QRDetector DepthCamera $${DepthSemanticModuleName} MiddleController DefaultTask AudioController Calibration VisionBag MemCatch ImuFusion Imu &
elif [ ${arr[@]:0:1} != "SR" ]; then
    echo "配置设备"
    ./ixRobot --with DataCenter Logger Driver Dsl Odom DeviceDriver Docking DockLocalization $${LocalizationModuleName} TrackNavi ObstacleDetectorV2 CloudProcessor Comm $${SlamModuleName}
    Laser Camera QRDetector DepthCamera $${DepthSemanticModuleName} MiddleController DefaultTask AudioController Calibration CMDetect MemCatch ImuFusion Imu &
elif [ ${arr[@]:0:1} != "SR" ]; then
    echo "配置设备"
    ./ixRobot --with DataCenter Logger Driver Dsl Odom DeviceDriver Docking DockLocalization Localization TrackNavi ObstacleDetectorV2 CloudProcessor Comm CommTripart $${SlamModuleName}
    Laser Camera QRDetector DepthCamera $${DepthSemanticModuleName} MiddleController DefaultTask AudioController Calibration MemCatch ImuFusion Imu &
elif [ "${robot_type}" = "SR-S-360-S-A2" ] || [ "${robot_type}" = "SR-S-360-S-A2" ] || [ "${robot_type}" = "SR-S-360-S-A2" ]; then
    echo "配置传感器"
    ./ixRobot --with DataCenter Logger Driver Dsl Odom DeviceDriver Docking Comm DefaultTask MemCatch MiddleController &
elif [ "${robot_type}" = "SR-S-360-S-A2" ]; then
    echo "配置传感器"
    ./ixRobot --with DataCenter Logger Driver Dsl Odom Docking Comm DefaultTask MemCatch MiddleController &
elif [ ${arr[@]:0:1} != "SR" ]; then
    echo "配置设备"
    ./ixRobot --with DataCenter Logger Driver Dsl Comm Laser DepthCamera MiddleController DefaultTask MemCatch VisionDock &
elif [ ${arr[@]:0:1} != "SR" ]; then
    export LD_LIBRARY_PATH=/home/f151b/hntal/lib/ake:/lib/ake:/LD_LIBRARY_PATH
    ./ixRobot --with DataCenter Logger Driver Dsl Odom DeviceDriver Docking DockLocalization $${LocalizationModuleName} TrackNavi ObstacleDetectorV2 CloudProcessor Comm $${SlamModuleName}
    Laser Camera QRDetector DepthCamera $${DepthSemanticModuleName} MiddleController RestRpc DefaultTask AudioController Calibration MemCatch ImuFusion Imu VisionDock &
elif [ "${robot_type}" = "Robot_Localization" ]; then
    echo "配置软件"
    ./ixRobot --with DataCenter Logger Odom RestRpc Comm Calibration MiddleController DefaultTask Laser $${LocalizationModuleName} $${SlamModuleName} &
else
    echo "配置设备"
    ./ixRobot --with DataCenter Logger Driver DriverDsl Odom DeviceDriver Docking DockLocalization $${LocalizationModuleName} TrackNavi ObstacleDetectorV2 CloudProcessor Comm $${SlamModuleName}
    Laser Camera QRDetector DepthCamera $${DepthSemanticModuleName} MiddleController DefaultTask AudioController Calibration MemCatch ImuFusion Imu Arm VisionDock &

```

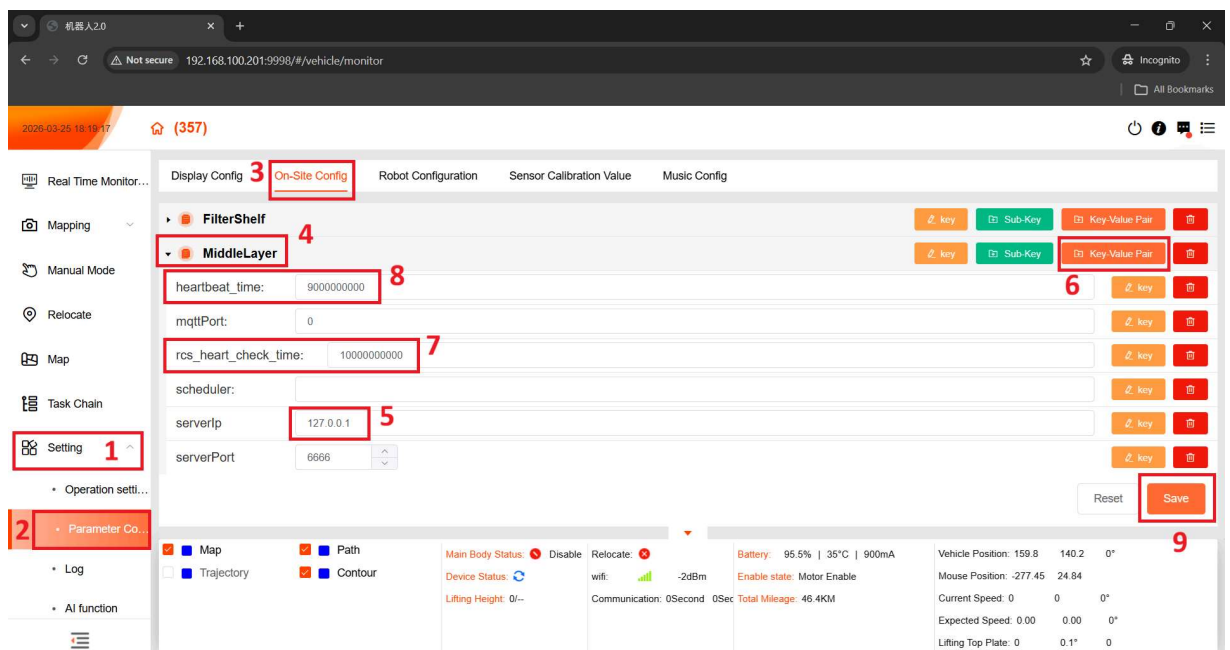
9. FileZilla upload the edited file in step #8 to **Overwrite** ~/hontai/start.sh. Make sure **file attributes** are **NOT changed**.

10. Open browser of Lanxin controller (<http://<NavWiz IP Address>:9998>) and restart device.



11. Update **MiddleLayer** parameters (refresh the browser if not shown)

- Change **调度ip** or **serverIp** to **127.0.0.1**
- Add **rcs_heart_check_time** with value **1000000000**
- Add **heartbeat_time** with value **9000000000**
- Make sure the **Save** button is clicked.



12. Restart device same as in step #10.

13. Soft Reboot NavWiz.